Intrusive Polynomial Chaos for CFD using OpenFOAM

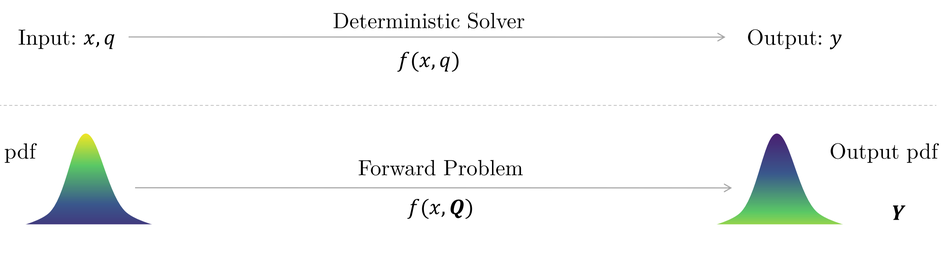
Jigar Parekh, Roel Verstappen
Bernoulli Institute, University of Groningen, Netherlands

university of groningen

Introduction

Motivation

- Large Eddy Simulations (LES) are known to provide detailed information about the dynamics of a turbulent flow.
- The accuracy of these simulation depend not only on the quality of the turbulence model but also on the associated uncertainties.
- Polynomial chaos representations allow us to quantify and propagate uncertainties in CFD simulations at a much lower cost than the standard Monte Carlo (MC) approach.



Uncertainty in
- physical or model parameters
- initial or boundary conditions

UQ Approach
- make use of deterministic solver
- develop surrogate model

Quantity of interest
- mean, variance of velocity
- spread of the wake

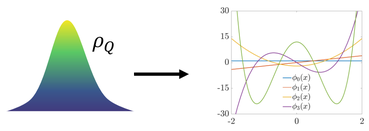
Objectives

- Develop an Intrusive Polynomial Chaos (IPC) solver for CFD using an existing deterministic solver in OpenFOAM.
- Uncertainty -Quantification (UQ) and -Propagation (UP): quantify the error bounds of the LES model and analyze their nonlinear propagation.
- Apply this solver to estimate the mean and the variance, as well as the confidence intervals for the quantities of interest.

Intrusive Polynomial Chaos for CFD – Idea & Implementation

Generalized Polynomial Chaos Expansion (gPCE)

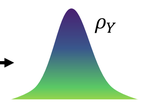
STEP 1.
Create polynomial chaoses (w.r.t. ρ_Q)



s.t. $\int \phi_i(q)\phi_j(q)\rho_Q(q)dq = \begin{cases}\gamma_i & i = j \\ 0 & i \neq j\end{cases}$

STEP 2.
Approximate r.v. using poly. chaos

$$Y = f(x,q) = \sum_{i=0}^{\infty} f_i(x)\phi_i(q), \quad \mu = f_0, \quad \sigma^2 = \sum_{i=1}^{\infty} f_i^2\phi_i^2$$



The problem reduces to finding the coefficients $f_i(x)$

✓ Non-intrusive Polynomial Chaos

$N_k = (n_k + 1)^d$ quad. points

$$f_i(x) \approx \frac{1}{\gamma_i}\sum_{k=1}^{N_k} f(q_k)\phi_i(q_k)w_k$$

Deterministic Solver

✓ **Intrusive Polynomial Chaos**

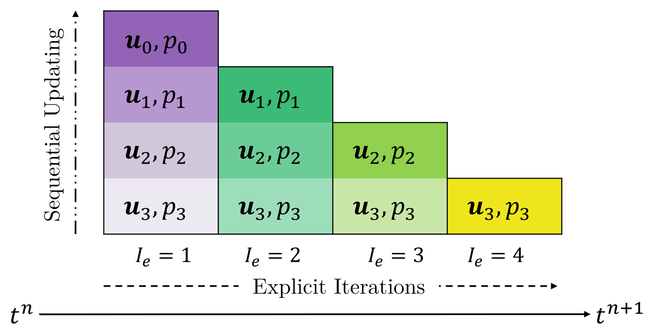
$Y \approx \sum_{i=0}^{P} f_i(x)\phi_i(q)$
$P + 1 = (n + d)!/(n!\, d!)$

Deterministic Solver

System of $(P + 1)$ PDEs $\quad \frac{\partial f_k(x)}{\partial x} = \Psi(x, \phi) + \ \ldots$

N_k, $(P + 1) \ll$ Monte Carlo (MC) Samples

Evolution of Mode Strengths

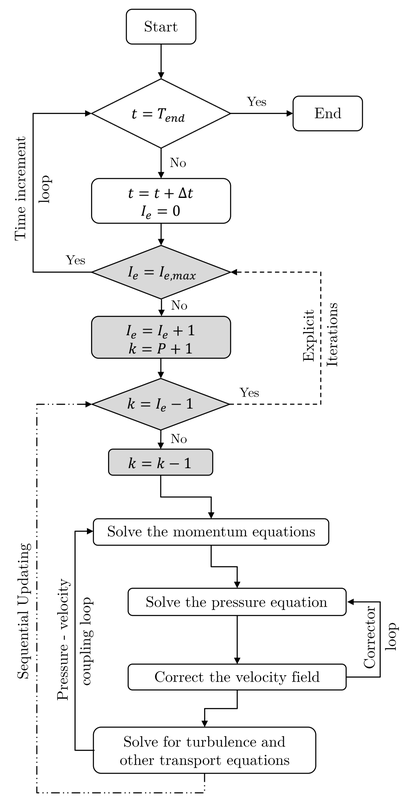


Stochastic Form of Filtered Naiver-Stokes with Smagorinsky LES Model

$$\boldsymbol{\nabla} \cdot \boldsymbol{u_k} = 0,$$

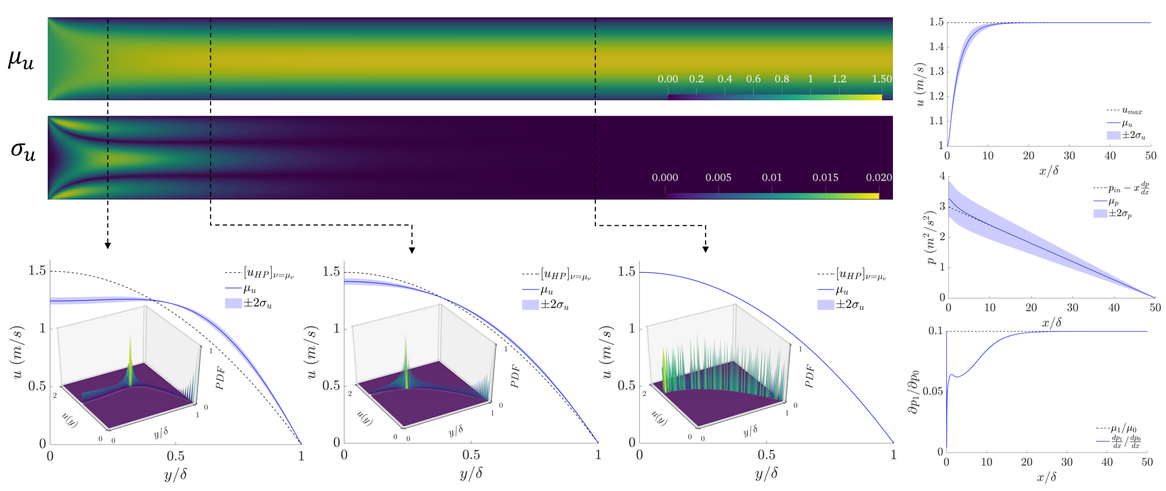
$$\frac{\partial \boldsymbol{u_k}}{\partial t} + \sum_{i=0}^{P}\sum_{j=0}^{P} (\boldsymbol{u_i}\cdot\boldsymbol{\nabla})\boldsymbol{u_j} C_{ijk}(\phi) = -\boldsymbol{\nabla} p_k + \sum_{i=0}^{P}\sum_{j=0}^{P} \boldsymbol{\nabla}\cdot(\nu_i\, \boldsymbol{\nabla}\, \boldsymbol{u_j}) C_{ijk}(\phi)$$
$$+ \sum_{i=0}^{P}\sum_{j=0}^{P}\sum_{l=0}^{P}\sum_{m=0}^{P} \boldsymbol{\nabla}\cdot(C_{s_l}C_{s_m}\,\Delta^2|\boldsymbol{S}|_i \boldsymbol{\nabla}\, \boldsymbol{u_j}) C_{ijklm}(\phi),$$

for $k = 0,1, \ldots, P$

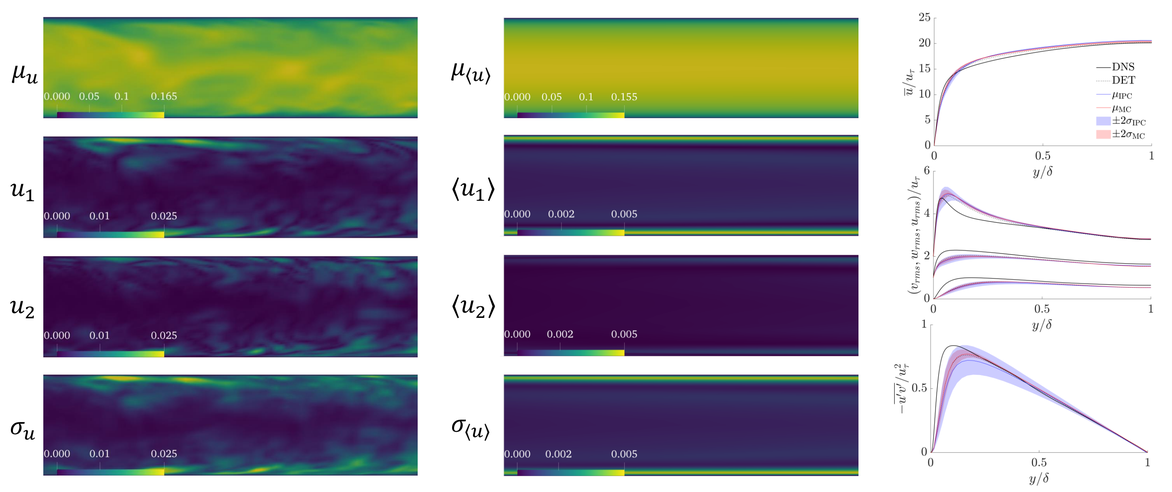


Test Cases & Application

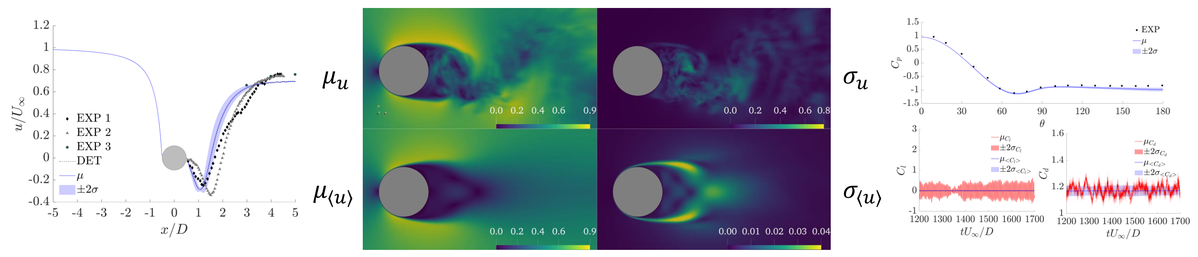
Plane Poiseuille Flow | $Re = 100$ | $\nu \sim \mathcal{N}(2e-3, 2e-4)$, CV = 10%



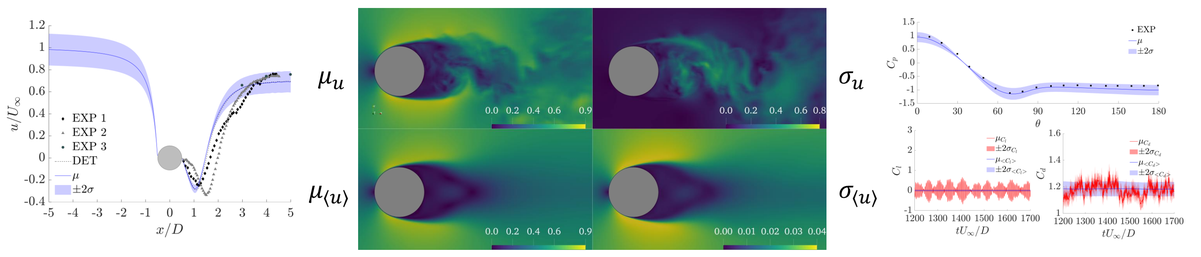
Turbulent Channel Flow | $Re_\tau = 395$ | $C_S \sim \mathcal{U}(0.075, 0.125)$, CV ≈ 14%

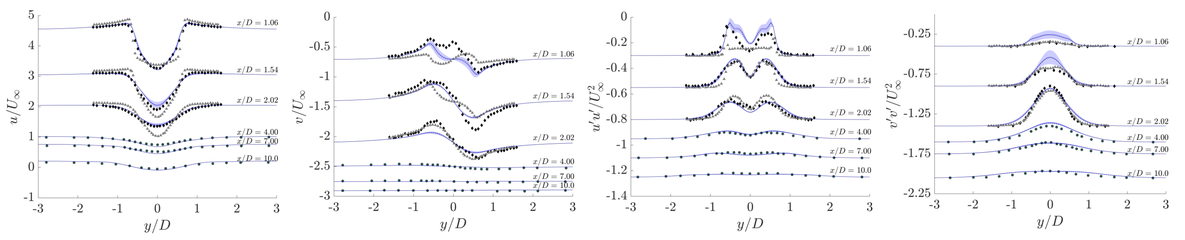


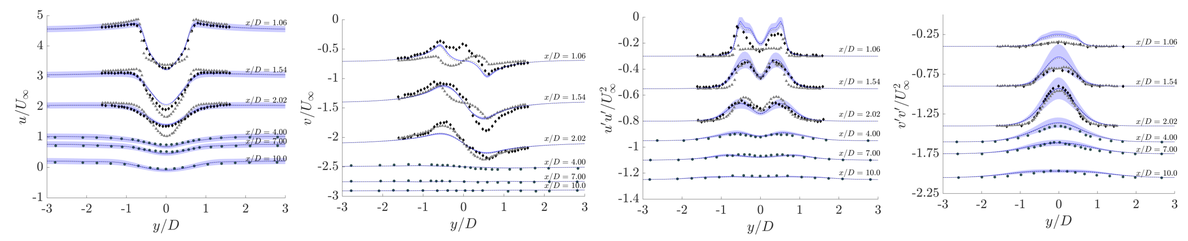
Flow Over a Cylinder | $Re = 3900$ | $C_S \sim \mathcal{U}(0.05, 0.08)$, CV ≈ 13%



Flow Over a Cylinder | $Re = 3900$ | $u_{inlet} \sim \mathcal{U}(0.54, 0.64)$, CV ≈ 5%







Conclusions & Future Work

Inferences

- Mean and variance obtained from IPC are in accordance with MC approach. Also, the estimated mean is close to the deterministic solution.
- The IPC solver is capable of capturing the influence of different forms of uncertainties on laminar and turbulent flow fields.



Forthcoming

- Application of the IPC solver for the uncertainty quantification and propagation in wind-farm simulations.
- Parallel in Time (PinT) method for faster IPC calculations, with a general framework of PinT-IPC for a system of PDEs.

Contact - j.parekh@rug.nl | GitHub - https://github.com/parallelwindfarms/gPCPimpleFoam